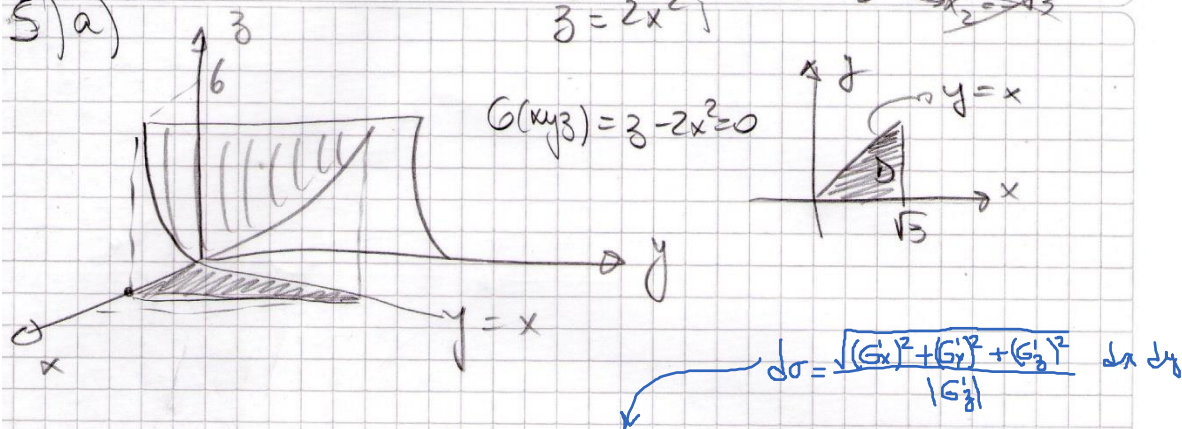


EJERCICIOS RESUELTOS TP 10

Calcule el área del trozo de superficie cilíndrica $z=2x^2$ con $y \leq x$, $z \leq 6$, 1er. oct.

5) a)



$$\text{Area } S = \iint_S d\sigma = \iint_D \frac{\sqrt{16x^2 + 1}}{|1|} dx dy =$$

$$= \iint_D \sqrt{16x^2 + 1} dx dy = \int_0^{\sqrt{3}} dx \int_0^x \sqrt{16x^2 + 1} dy =$$

$$= \int_0^{\sqrt{3}} x \sqrt{16x^2 + 1} dx = \frac{1}{48} (16x^2 + 1)^{3/2} \Big|_0^{\sqrt{3}} =$$

$$\int x \sqrt{16x^2 + 1} dx = \int \cancel{x} \sqrt{u} \frac{du}{32\cancel{x}} = \frac{1}{32} \frac{u^{3/2}}{3/2} =$$

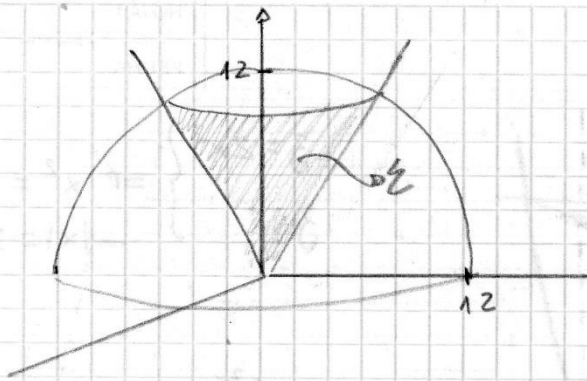
$$u = 16x^2 + 1$$

$$du = 32x dx \Rightarrow dx = \frac{du}{32x} = \frac{1}{32} \frac{du}{x} = \frac{1}{32} \frac{du}{\frac{\sqrt{u}}{4}} = \frac{1}{8} \frac{du}{\sqrt{u}} = \frac{1}{8} \frac{du}{u^{1/2}} = \frac{1}{8} \cdot \frac{2}{1/2} u^{1/2} = \frac{1}{4} u^{1/2}$$

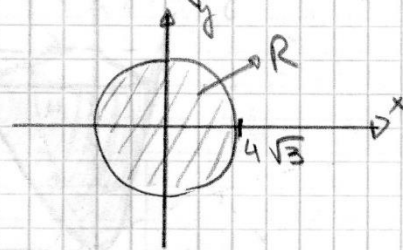
$$= \frac{1}{48} (16 \cdot 3 + 1)^{3/2} - \frac{1}{48} \cdot 1^{3/2} = \frac{1}{48} \cdot 49^{3/2} - \frac{1}{48} =$$

$$= \frac{343}{48} - \frac{1}{48} = \frac{342}{48} = \frac{57}{8}$$

b)



$$\begin{cases} x^2 + y^2 + z^2 = 12^2 \\ z^2 = 2x^2 + 2y^2 \end{cases} \Rightarrow$$



$$\Rightarrow x^2 + y^2 + 2x^2 + 2y^2 = 12^2 \Rightarrow 3(x^2 + y^2) = 12^2 \Rightarrow$$

$$\Rightarrow x^2 + y^2 = 12 \cdot 4 \Rightarrow x^2 + y^2 = (4\sqrt{3})^2$$

$$\text{Area } \varepsilon = \iint_{\varepsilon} d\sigma = \iint_R \frac{\sqrt{F_x'^2 + F_y'^2 + F_z'^2}}{|F_z'|} dx dy = (*)$$

$$F(x, y, z) = z^2 - 2x^2 - 2y^2 = 0$$

$$(*) = \iint_R \frac{\sqrt{(-4x)^2 + (-4y)^2 + (2z)^2}}{2z} dx dy = \iint_R \frac{\sqrt{16x^2 + 16y^2 + 4z^2}}{2z} dx dy$$

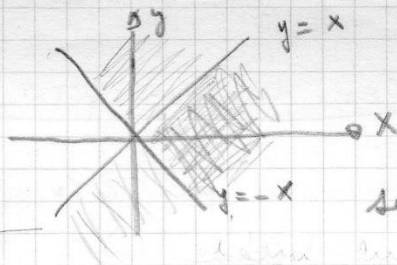
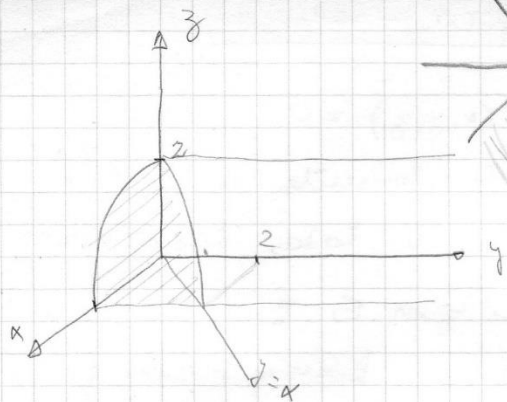
$$= \iint_R \frac{\sqrt{8(2x^2 + 2y^2) + 4z^2}}{2z} dx dy = \iint_R \frac{\sqrt{16x^2 + 16y^2 + 8x^2 + 8y^2}}{2\sqrt{2x^2 + 2y^2}} dx dy$$

$$= \iint_R \frac{\sqrt{24} \sqrt{x^2 + y^2}}{2\sqrt{2} \sqrt{x^2 + y^2}} dx dy = \sqrt{3} \iint_R dx dy =$$

$$= \sqrt{3} \int_0^{2\pi} d\phi \int_0^{4\sqrt{3}} \rho d\rho = \sqrt{3} 2\pi \left[\frac{\rho^2}{2} \right]_0^{4\sqrt{3}} =$$

$$= \frac{\sqrt{3} 2\pi}{2} 16 \cdot 3 = 48\sqrt{3} \pi$$

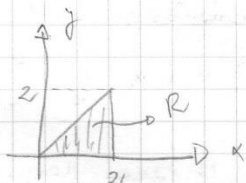
05)
C)



proporção (xy) de
 $-x \leq y \leq x$

sólo $x \geq 0$

idem outro octante com $x \geq 0$



$$A_1 = \iint_S ds = \iint_R \frac{\sqrt{(F'_x)^2 + (F'_y)^2 + (F'_z)^2}}{|F'_z|} dx dy = \textcircled{1}$$

$$F(x, y, z) = x^2 + z^2 - 4 = 0$$

$$\textcircled{1} = \iint_R \frac{\sqrt{4x^2 + 0 + 4z^2}}{|2z|} dx dy = \iint_R \frac{2\sqrt{x^2 + z^2}}{2\sqrt{4-x^2}} dx dy$$

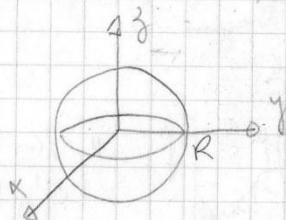
$$\textcircled{2} = \iint_R \frac{\sqrt{4}}{\sqrt{4-x^2}} = 2 \iint_R \frac{dx dy}{\sqrt{4-x^2}} =$$

$$= 2 \int_0^2 dx \int_0^x dy \frac{1}{\sqrt{4-x^2}} = 2 \int_0^2 dx \frac{x}{\sqrt{4-x^2}} =$$

$$= 2 \left[-\sqrt{4-x^2} \right]_0^2 = 2(-\sqrt{0} + \sqrt{4}) = 2 \cdot 2 = 4$$

$$\text{Área total} = 2 \cdot A_1 = 8$$

d)



$$\vec{r}(\theta, \lambda) = (R \cos \theta \sin \lambda, R \sin \theta \sin \lambda, R \cos \lambda)$$

$$0 \leq \theta \leq 2\pi \wedge 0 \leq \lambda \leq \pi$$

$$\vec{n} = \frac{\partial \vec{r}}{\partial \theta} \wedge \frac{\partial \vec{r}}{\partial \lambda} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -R \sin \theta \sin \lambda & R \cos \theta \sin \lambda & 0 \\ R \cos \theta \cos \lambda & R \sin \theta \cos \lambda & -R \sin \lambda \end{vmatrix} =$$

$$= \vec{i} (-R^2 \cos \theta \sin^2 \lambda) - \vec{j} (R^2 \sin \theta \sin^2 \lambda) + \vec{k} (R^2 \sin^2 \theta \cdot \sin \lambda \cos \lambda - R^2 \cos^2 \theta \cos \lambda \sin \lambda) =$$

$$= (-R^2 \cos \theta \sin^2 \lambda, -R^2 \sin \theta \sin^2 \lambda, R^2 \sin \lambda \cos \lambda)$$

$$|\vec{n}| = \sqrt{R^4 \cos^2 \theta \sin^4 \lambda + R^4 \sin^2 \theta \sin^4 \lambda + R^2 \sin^2 \lambda \cos^2 \lambda} =$$

$$= R^2 \sqrt{\sin^4 \lambda (\cos^2 \theta + \sin^2 \theta) + \sin^2 \lambda \cos^2 \lambda} =$$

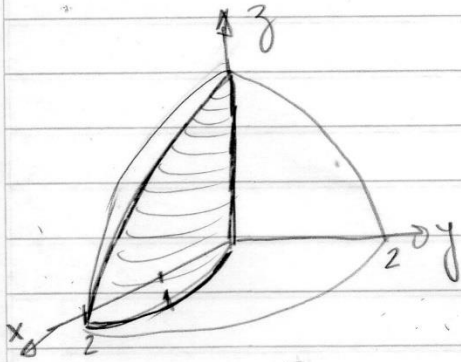
$$= R^2 \sqrt{\sin^2 \lambda (\sin^2 \lambda + \cos^2 \lambda)} = R^2 |\sin \lambda| =$$

$$= R^2 \sin \lambda \quad (\text{pourquoi } 0 \leq \lambda \leq \pi \Rightarrow \sin \lambda \geq 0)$$

$$\text{Area } S = \iint_S d\sigma = \int_0^{2\pi} d\theta \int_0^\pi d\lambda \, R^2 \sin \lambda =$$

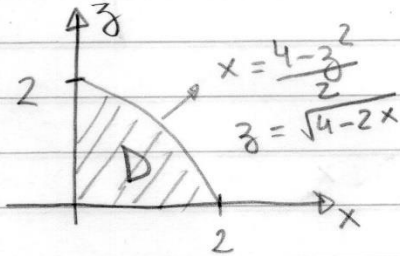
$$= R^2 \cdot 2\pi \cdot [-\cos \lambda]_0^\pi = 2\pi R^2 (1 + 1) = 4\pi R^2$$

s)e) $x^2 + y^2 = 2x$ con $x^2 + y^2 + z^2 \leq 4$; 1° ord



$$\begin{aligned} x^2 + y^2 &= 2x \\ x^2 - 2x + 1 + y^2 &= 1 \\ (x-1)^2 + y^2 &= 1 \end{aligned}$$

$$F(x, y, z) = x^2 + y^2 - 2x = 0$$



$$\left. \begin{aligned} x^2 + y^2 &= 2x \\ x^2 + y^2 + z^2 &= 4 \end{aligned} \right\} \Rightarrow 0$$

$$\Rightarrow 2x + z^2 = 4 \Rightarrow x = \frac{4 - z^2}{2}$$

$$z = \sqrt{4 - 2x}$$

$$\text{Area } S = \iint_S d\sigma = \iint_D \frac{\sqrt{(F'_x)^2 + (F'_y)^2 + (F'_z)^2}}{|F'_y|} dx dz =$$

$$= \iint_D \frac{\sqrt{(2x-2)^2 + 4y^2}}{|2y|} dx dz = \iint_D \frac{2\sqrt{(x-1)^2 + y^2}}{2y} dx dz =$$

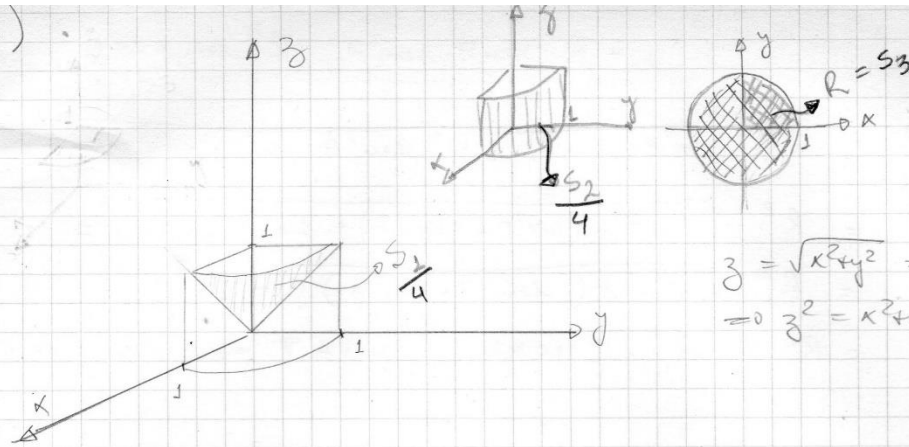
$$= \iint_D \frac{\sqrt{x^2 - 2x + 1 + 2x - x^2}}{\sqrt{2x - x^2}} dx dz = \iint_D \frac{1}{\sqrt{2x - x^2}} dx dz =$$

$$= \int_0^2 dx \int_0^{\sqrt{4-2x}} \frac{1}{\sqrt{2x-x^2}} dz = \int_0^2 \frac{\sqrt{4-2x}}{\sqrt{2x-x^2}} dx =$$

$$= \int_0^2 \frac{\sqrt{2(2-x)}}{\sqrt{x(2-x)}} dx = \sqrt{2} \int_0^2 \frac{1}{\sqrt{x}} dx = \sqrt{2} \left[2\sqrt{x} \Big|_0^2 \right] =$$

$$= \sqrt{2} \cdot 2\sqrt{2} = \textcircled{4}$$

f)



$$z = \sqrt{x^2 + y^2} \Rightarrow z^2 = x^2 + y^2$$

S₁

$$\text{Area } S_1 = \iint_{S_1} dS_1 = \iint_R \frac{\sqrt{4x^2 + 4y^2 + 4z^2}}{|1 - 2z|} dx dy = \iint_R \frac{z \sqrt{x^2 + y^2 + z^2}}{z \sqrt{x^2 + y^2}} dx dy = \textcircled{1}$$

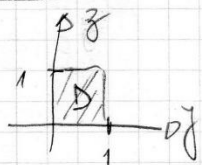
$$F_1(x, y, z) = x^2 + y^2 - z^2 = 0$$

$$\textcircled{1} = \iint_R \frac{\sqrt{x^2 + y^2 + x^2 + y^2}}{\sqrt{x^2 + y^2}} dx dy = \iint_R \sqrt{2} \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}} dx dy =$$

$$= \int_0^{2\pi} d\theta \int_0^1 d\rho \rho \sqrt{2} = \sqrt{2} \cdot \frac{2\pi}{2} \left[\frac{\rho^2}{2} \right]_0^1 =$$

$$= \sqrt{2} 2\pi \left(\frac{1}{2} - 0 \right) = \sqrt{2} \pi$$

$S_2 \mid x^2 + y^2 = 1 \rightarrow G(x, y, z) = x^2 + y^2 - 1 = 0$



$$\frac{\text{Area } S_2}{4} = \iint_{S_2} d\sigma = \iint_D \frac{\sqrt{4x^2 + 4y^2}}{|2x|} dy dz =$$

$$= \iint_D \frac{\sqrt{x^2 + y^2}}{x} dy dz =$$

$$= \iint_D \frac{\sqrt{1 - y^2 + y^2}}{\sqrt{1 - y^2}} dy dz = \int_0^1 dy \int_0^1 dz \frac{1}{\sqrt{1 - y^2}} =$$

$$= \int_0^1 \frac{1}{\sqrt{1 - y^2}} dy = \arcsin y \Big|_0^1 = \pi/2 = \frac{\text{Area } S_2}{4} \Rightarrow$$

$$\Rightarrow \text{Area } S_2 = 2\pi$$

$S_3 \mid \text{Area } S_3 \rightarrow \text{de círculo de radio } 1$

$$\text{Area } S_3 = \pi \cdot 1^2 = \pi$$

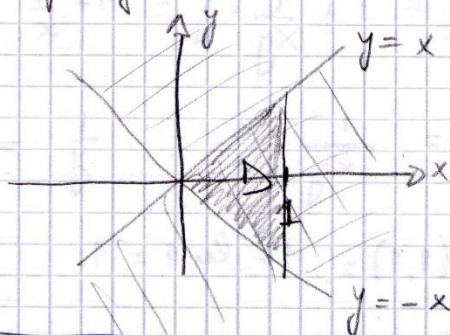
$$\text{Area total } S = \text{Area } S_1 + \text{Area } S_2 + \text{Area } S_3$$

$$= \sqrt{2}\pi + 2\pi + \pi = (\sqrt{2} + 3)\pi$$

5) g) $\text{sup } z = x^2 - y \rightarrow$ difícil de graficar

se puede graficar la proyección en el $pl(xy)$

$$\underbrace{|y| < x, x < 1}_{-x < y < x}$$



$$F(x, y, z) = x^2 - y - z = 0$$

$$\text{Area } S = \iint_D \cdot d\sigma = \iint_D \frac{\sqrt{4x^2 + 1 + 1}}{|-1|} dx dy =$$

$$= \iint_D \sqrt{4x^2 + 2} dx dy = \int_0^1 dx \int_{-x}^x \sqrt{4x^2 + 2} dy =$$

$$= \int_0^1 dx \quad 2x \sqrt{4x^2 + 2} = \left[(4x^2 + 2)^{3/2} \cdot \frac{1}{4} \cdot \frac{2}{3} \right]_0^1 =$$

$$= \frac{1}{6} \left[6^{3/2} - 2^{3/2} \right] = \frac{1}{6} (6\sqrt{6} - 2\sqrt{2}) =$$

$$= \sqrt{6} - \frac{\sqrt{2}}{3}$$

5) h) $z = x + \ln(xy)$ en $(1,1,z_0)$

$$z'_x = 1 + \frac{1}{xy} \cdot y = 1 + \frac{1}{x} \Rightarrow z'_x(1,1) = 2$$

$$z'_y = \frac{1}{xy} \cdot x = \frac{1}{y} \Rightarrow z'_y(1,1) = 1$$

$$z_0(1,1) = 1 + \ln 1 = 1$$

Pl. tange

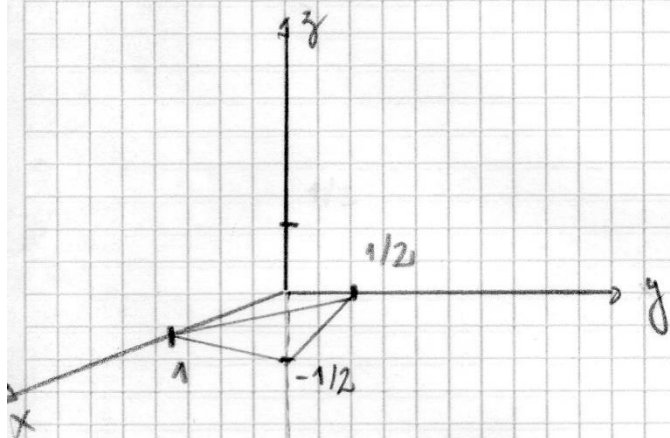
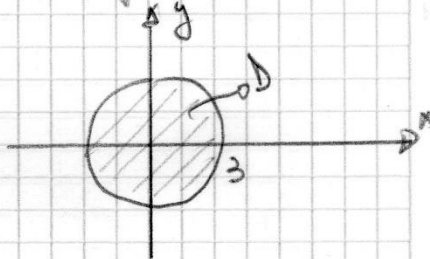
$$z - 1 = 2(x - 1) + 1(y - 1)$$

$$\cancel{z} - \cancel{1} = 2x - 2 + y - 1$$

$$2x + y - z = 2$$

$$F(x,y,z) = 2x + y - z - 2 = 0$$

Proyeksi en $pl(xy)$

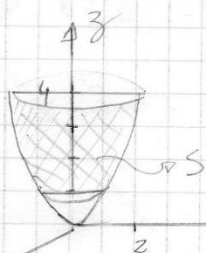


$$\text{Area } S = \iint_{\Sigma} d\sigma = \iint_D \frac{\sqrt{F_x'^2 + F_y'^2 + F_z'^2}}{|F_z'|} dx dy =$$

$$= \iint_D \frac{\sqrt{2^2 + 1^2 + 1^2}}{1} dx dy = \sqrt{6} \int_0^{2\pi} d\sigma \int_0^3 \rho d\rho =$$

$$= \sqrt{6} 2\pi \cdot \left[\frac{\rho^2}{2} \Big|_0^3 \right] = 2\sqrt{6} \pi \cdot \frac{9}{2} = 9\sqrt{6} \pi$$

06)



$$\begin{cases} z = x^2 + y^2 \\ z = 1 \end{cases} \Rightarrow x^2 + y^2 = 1$$

$$\begin{cases} z = x^2 + y^2 \\ z = 4 \end{cases} \Rightarrow x^2 + y^2 = 4$$

sólo el frente $\rightarrow x \geq 0$

$$F(x, y, z) = x^2 + y^2 - z = 0$$

$$I_3 = \int_S (x^2 + y^2) \delta(x, y, z) d\sigma = \int_S (x^2 + y^2) \frac{k}{x^2 + y^2} d\sigma =$$

$$= k \iint_D \frac{\sqrt{4x^2 + 4y^2 + 1}}{1-1} dx dy = k \iint_D \sqrt{4(x^2 + y^2) + 1} dx dy =$$

$$= k \int_{-\pi/2}^{\pi/2} d\theta \int_1^2 \rho \sqrt{4\rho^2 + 1} d\rho = k \int_{-\pi/2}^{\pi/2} d\theta \left[\frac{(4\rho^2 + 1)^{3/2}}{3} \right]_1^2 =$$

$$= k \int_{-\pi/2}^{\pi/2} d\theta \left[\frac{1}{12} (17^{3/2} - 5^{3/2}) \right] = k \cdot \frac{\pi}{2} \cdot \frac{1}{12} (17^{3/2} - 5^{3/2}) =$$

$$= \frac{k\pi}{12} (17^{3/2} - 5^{3/2})$$

$$7) \iint_S \vec{F} \cdot \vec{n} \, d\sigma = \iint_S (k \vec{n}) \cdot \vec{n} \, d\sigma = k \iint_S \|\vec{n}\|^2 \, d\sigma =$$

$$= k \iint_S d\sigma \quad (*) \quad (\text{porque } \|\vec{n}\| = 1)$$

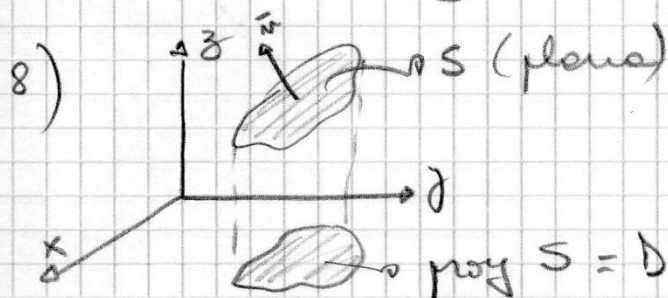
Si $\vec{F} = k \vec{n} \Rightarrow \|\vec{F}\| = \|k \vec{n}\| = |k| \|\vec{n}\| =$

$$= k \cdot 1 = k \quad (\text{porque } k > 0)$$

Luego $k = \|\vec{F}\|$ - reemplazando en $(*)$:

$$k \iint_S d\sigma = \|\vec{F}\| \underbrace{\iint_S d\sigma}_{\text{área } S} = \|\vec{F}\| \text{ área } S$$

Conclusión: $\iint_S \vec{F} \cdot \vec{n} \, d\sigma = \|\vec{F}\| \text{ área } S$

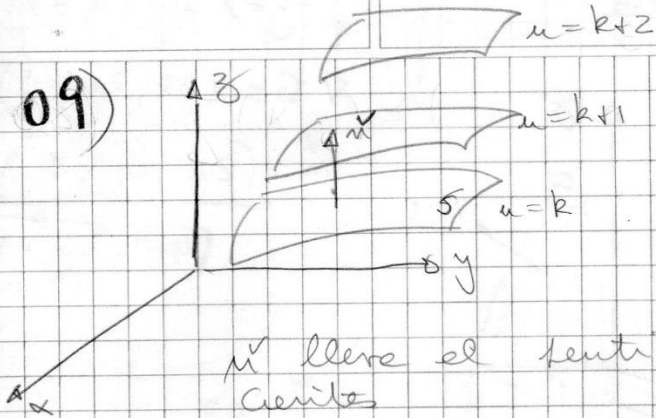


$\vec{F} = \vec{k}$ (constante)

$$\iint_S \vec{F} \cdot \vec{n} \, d\sigma = \iint_S \underbrace{\vec{k} \cdot \vec{n}}_{\text{n° real}} \, d\sigma = (\vec{k} \cdot \vec{n}) \underbrace{\iint_S d\sigma}_{\text{área } S} =$$

$$= (\vec{k} \cdot \vec{n}) \text{ área } S = (\vec{F} \cdot \vec{n}) (\text{área } S)$$

09)



$$\vec{T} = \nabla \phi \in C^1$$

$$S \cap \phi = k$$

$$\vec{n} = \frac{\nabla \phi}{\|\nabla \phi\|}$$

(porque el grad. apunta hacia donde la k crece)

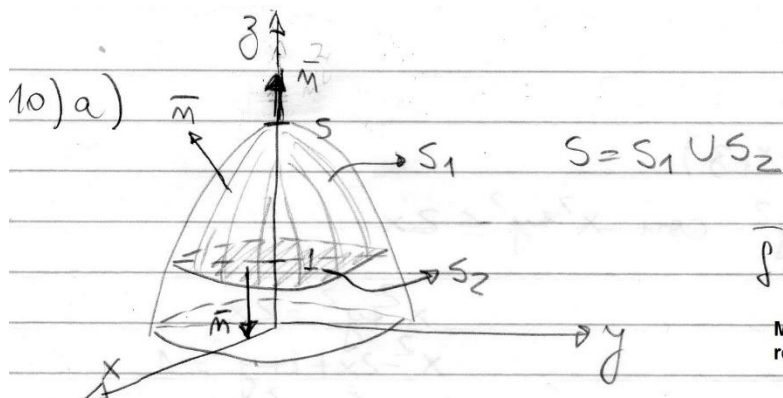
$\vec{n}$ llena el sentido de los potenciales crecientes

$$\iint_S (\vec{n} \cdot \vec{n}) \, ds = \iint_S (\nabla \phi \cdot \vec{n}) \, ds = \iint_S \nabla \phi \cdot \frac{\nabla \phi}{\|\nabla \phi\|} \, ds$$

$$= \iint_S \frac{(\nabla \phi)^2}{\|\nabla \phi\|} \, ds = \iint_S \frac{\|\nabla \phi\|^2}{\|\nabla \phi\|} \, ds =$$

$$= \iint_S \underbrace{\|\nabla \phi\|}_{>0} \, ds > 0$$

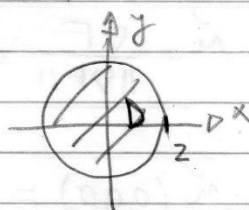
$\times$ es una norma



$$\iint_S \vec{F} \cdot \vec{n} \, d\sigma = \iint_{S_1} \vec{F} \cdot \vec{n} \, d\sigma + \iint_{S_2} \vec{F} \cdot \vec{n} \, d\sigma$$

S_1 $z = 5 - x^2 - y^2$ con $z \geq 1$

$$\begin{cases} z = 5 - x^2 - y^2 \\ z = 1 \end{cases} \Rightarrow x^2 + y^2 = 4$$



$$F(x, y, z) = z + x^2 + y^2 - 5 = 0$$

$$\vec{n} = \frac{(2x, 2y, 1)}{\sqrt{4x^2 + 4y^2 + 1}} \quad \vec{n}(0, 0, 5) = (0, 0, 1)$$

$$\iint_{S_1} \vec{F} \cdot \vec{n} \, d\sigma = \iint_{S_1} (x, y, z) \cdot \frac{(2x, 2y, 1)}{\sqrt{4x^2 + 4y^2 + 1}} \, d\sigma =$$

$$= \iint_{S_1} \frac{2x^2 + 2y^2 + z}{\sqrt{4x^2 + 4y^2 + 1}} \, d\sigma =$$

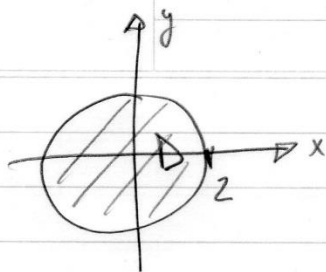
$$= \iint_D \frac{2x^2 + 2y^2 + z}{\sqrt{4x^2 + 4y^2 + 1}} \cdot \frac{\sqrt{4x^2 + 4y^2 + 1}}{1} \, dx \, dy =$$

$$= \iint_D (2x^2 + 2y^2 + 5 - x^2 - y^2) \, dx \, dy = \iint_D (5 + x^2 + y^2) \, dx \, dy =$$

$$= \int_0^{2\pi} d\theta \int_0^2 dp (5 + p^2) \cdot p = 2\pi \cdot \left[\frac{5p^2}{2} + \frac{p^4}{4} \right]_0^2 =$$

$$= 2\pi [10 + 4] = 28\pi$$

S_2



$$\begin{cases} z = 5 - x^2 - y^2 \\ z = 1 \end{cases} \Rightarrow x^2 + y^2 = 4$$

$$S_2 \rightarrow z = 1 \quad \vec{n} = (0, 0, -1)$$

porque debe ser orientada exterior, en concordancia con la orientación de S_1)

$$G(x, y, z) = z - 1 = 0$$

$$\iint_{S_2} \vec{F} \cdot \vec{n} \, d\sigma = \iint_{S_2} (x, y, z) \cdot (0, 0, -1) \, d\sigma =$$

$$= \iint_{S_2} -z \, d\sigma = \iint_D -z \frac{\sqrt{0^2 + 0^2 + 1}}{|1|} \, dx \, dy =$$

$$= 1$$

$$= - \iint_D \underbrace{1}_{z=1} \, dx \, dy = - \underbrace{\int_0^{2\pi} d\theta \int_0^2 dp \cdot p}_{\text{área del círculo}} = -\pi \cdot 4$$

$$\iint_S \vec{F} \cdot \vec{n} \, d\sigma = \underbrace{28\pi}_{S_1} - \underbrace{4\pi}_{S_2}$$

Observación: si la proyección no deforma la superficie (como en el caso de S_2)

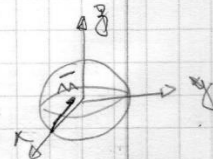
resulta

$$d\sigma = 1 \cdot dx \, dy$$

$$\frac{\sqrt{(F'_x)^2 + (F'_y)^2 + (F'_z)^2}}{|F'_z|} = 1$$

$$10) b) \vec{g}(\theta, \lambda) = (2 \cos \theta \sin \lambda, 2 \sin \theta \sin \lambda, 2 \cos \lambda)$$

$$\begin{cases} 0 \leq \theta \leq 2\pi \\ 0 \leq \lambda \leq \pi \end{cases}$$



$$\iint_S \vec{g} \cdot \vec{n} \, d\sigma = \iint_D \vec{g} \cdot \frac{\frac{\partial \vec{g}}{\partial \theta} \times \frac{\partial \vec{g}}{\partial \lambda}}{\|\frac{\partial \vec{g}}{\partial \theta} \times \frac{\partial \vec{g}}{\partial \lambda}\|} \, d\theta \, d\lambda = \textcircled{1}$$

$$\frac{\partial \vec{g}}{\partial \theta} \times \frac{\partial \vec{g}}{\partial \lambda} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2 \sin \theta \sin \lambda & 2 \cos \theta \sin \lambda & 0 \\ 2 \cos \theta \cos \lambda & 2 \sin \theta \cos \lambda & -2 \sin \lambda \end{vmatrix} =$$

$$= \vec{i} (-4 \cos \theta \sin^2 \lambda) - \vec{j} (4 \sin \theta \sin^2 \lambda) + \vec{k} (-4 \sin^2 \theta \cos \lambda \sin \lambda - 4 \cos^2 \theta \sin \lambda \cos \lambda) =$$

$$= (-4 \cos \theta \sin^2 \lambda, -4 \sin \theta \sin^2 \lambda, -4 \sin \lambda \cos \lambda) \rightarrow \vec{n}$$

$$\vec{n}(0, \frac{\pi}{2}) = (-4, 0, 0) \quad \vec{g}(0, \frac{\pi}{2}) = (2, 0, 0) \quad \vec{n} \text{ entrante}$$

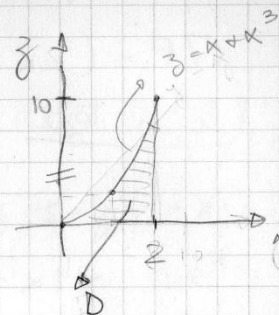
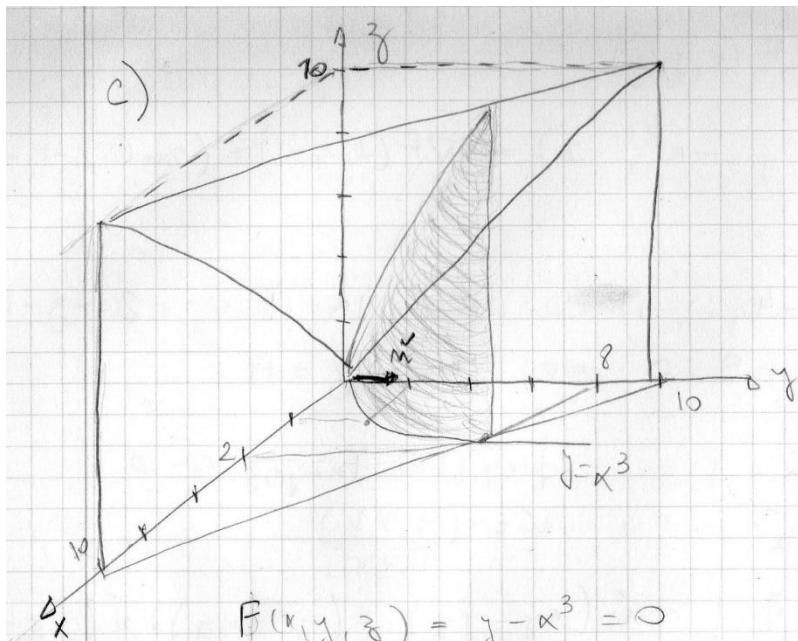
$$\textcircled{1} = \iint_D (2 \cos \theta \sin \lambda, 2 \sin \theta \sin \lambda, 2 \cos \lambda) \cdot (-4 \cos \theta \sin^2 \lambda,$$

$$-4 \sin \theta \sin^2 \lambda, -4 \sin \lambda \cos \lambda) \, d\theta \, d\lambda =$$

$$= \iint_D (-8 \cos^2 \theta \sin^3 \lambda - 8 \sin^2 \theta \sin^3 \lambda - 8 \cos^2 \lambda \sin \lambda) \, d\theta \, d\lambda =$$

$$= -8 \iint_D (\sin^3 \lambda + \cos^2 \lambda \sin \lambda) \, d\theta \, d\lambda = -8 \int_0^{2\pi} d\theta \int_0^{\pi} (\sin^3 \lambda + \cos^2 \lambda \sin \lambda) \, d\lambda =$$

$$= -8 \cdot 2\pi \left[-\cos \lambda + \frac{1}{3} \cos^3 \lambda - \frac{\cos^3 \lambda}{3} \right]_0^{\pi} = -16\pi [1 + 1] = -32\pi$$



$$\begin{cases} y = x^3 \\ z = x + y \end{cases} \Rightarrow \begin{cases} z = x + x^3 \\ x = 0 \rightarrow z = 0 \\ x = 2 \rightarrow z = 10 \\ x = 1 \rightarrow z = 2 \end{cases}$$

$$F(x, y, z) = y - x^3 = 0$$

$$\vec{n} = \frac{\nabla F}{\|\nabla F\|} = \frac{(-3x^2, 1, 0)}{\sqrt{9x^4 + 1}}$$

$$\vec{n}(0,0,0) = (0, 1, 0)$$

$$\iint_S \vec{r} \cdot \vec{n} \, d\sigma = \iint_S (xy, z, y - xz) \cdot \frac{(-3x^2, 1, 0)}{\sqrt{9x^4 + 1}} \, d\sigma =$$

$$\begin{cases} x + y = 10 \\ y = x^3 \end{cases} \Rightarrow 10 - x = x^3 \Rightarrow x^3 + x = 10$$

$$= \iint_S \frac{(-3x^3y + 3x)}{\sqrt{9x^4 + 1}} d\sigma = \iint_D \frac{(-3x^3y + 3x)}{\sqrt{9x^4 + 1}} \frac{\sqrt{9x^4 + 1}}{11} dx dz$$

$$= \iint_D (-3x^3y + 3x) dx dz = \iint_R \underbrace{(-3x^3x^3 + 3x)}_y dz dx =$$

$$= \int_0^2 dx \int_0^{x+x^3} dz (-3x^6 + 3x) = \int_0^2 dx \left[-3x^6 z + \frac{3z^2}{2} x \right]_0^{x+x^3} =$$

$$= \int_0^2 \left[-3x^6 (x+x^3) + \frac{(x+x^3)^2}{2} x \right] dx = \int_0^2 \left[-3x^7 - 3x^9 + \right. \\ \left. + \left(\frac{x^2}{2} + \frac{x^6}{2} + x^4 \right) x \right] dx = \int_0^2 \left(-3x^7 - 3x^9 + \frac{x^3}{2} + \frac{x^7}{2} + x^5 \right) dx =$$

$$= \int_0^2 \left(-3x^9 - \frac{5}{2}x^7 + x^5 + \frac{x^3}{2} \right) dx = \left[-\frac{3x^{10}}{10} - \frac{5}{2} \frac{x^8}{8} + \frac{x^6}{6} + \frac{x^4}{8} \right]_0^2$$

$$= -\frac{3 \cdot 2^{10}}{10} - \frac{5 \cdot 2^8}{2 \cdot 8} + \frac{2^6}{6} + \frac{2^4}{8} = -\frac{3072}{10} - \frac{1280}{16} + \frac{64}{6} + 2 =$$

$$= -\frac{5618}{15}$$

$$d) 0 = x^2 - yx^3 - z = F(x, y, z)$$

$$\nabla F = (2x - 3x^2y, -x^3, -1) = 0 \quad \nabla F(1, 2, -1) = (2-6, -1, -1)$$

Plane tangent

$$(x-1, y-2, z+1) \cdot (-4, -1, -1) = 0 \Rightarrow -4x + 4 - y + 2 - z - 1 = 0 \\ \Rightarrow -4x - y - z + 5 = 0 \Rightarrow 4x + y + z = 5$$

$$\Rightarrow z = 5 - 4x - y$$

$$\phi(x, y, z) = 4x + y + z - 5 = 0$$

$$\vec{n} = \frac{(4, 1, 1)}{\sqrt{16+1+1}}$$

$$\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ y & x & y \\ x & z & z \end{vmatrix} = \vec{i}(xy - yz) - \vec{j}(y^2 - xz) + \vec{k}(yz - x^2)$$

$$\vec{F} = (xy - yz, xy - y^2, yz - x^2)$$

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \iint_S (xy - yz, xy - y^2, yz - x^2) \cdot \frac{(4, 1, 1)}{\sqrt{18}} \, dS =$$

$$= \frac{1}{\sqrt{18}} \iint_R (4xy - 4yz + xy - y^2 + yz - x^2) \cdot \frac{\sqrt{18}}{|1|} \, dx \, dy$$

$$= \int_0^2 dx \int_1^3 dy [4xy - 4y(5-4x-y) + xy - y^2 + (5-4x-y)y - x^2]$$

$$= \int_0^2 dx \int_1^3 [4xy - 20y + 16xy + 4y^2 + xy - y^2 + 5y - 4xy - y^2 - x^2] \, dy =$$

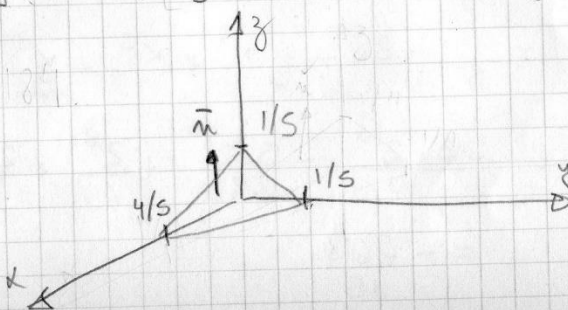
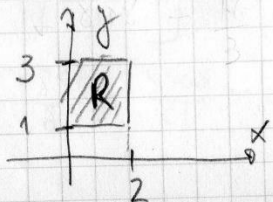
$$= \int_0^2 dx \int_1^3 (-15y + 17xy + 2y^2 - x^2) \, dy = \int_0^2 dx \left[-\frac{15y^2}{2} + \frac{17x y^2}{2} + \right.$$

$$\left. + \frac{2y^3}{3} - x^2 y \right]_1^3 = \int_0^2 dx \left(-\frac{15 \cdot 9}{2} + \frac{17x \cdot 9}{2} + \frac{2 \cdot 27}{3} - 3x^2 + \right.$$

$$\left(\frac{15}{2} - \frac{17}{2}x - \frac{2}{3} + x^2 \right) = \int_0^2 \left(-\frac{128}{3} + 68x - 2x^2 \right) dx =$$

$$= \left[-\frac{128}{3}x + \frac{68x^2}{2} - \frac{2x^3}{3} \right]_0^2 = -\frac{128 \cdot 2}{3} + \frac{68 \cdot 4}{2} - \frac{2 \cdot 8}{3} =$$

$$= \frac{136}{3}$$

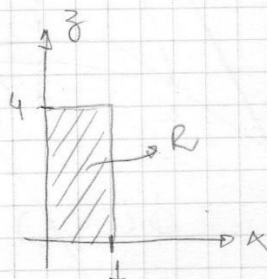
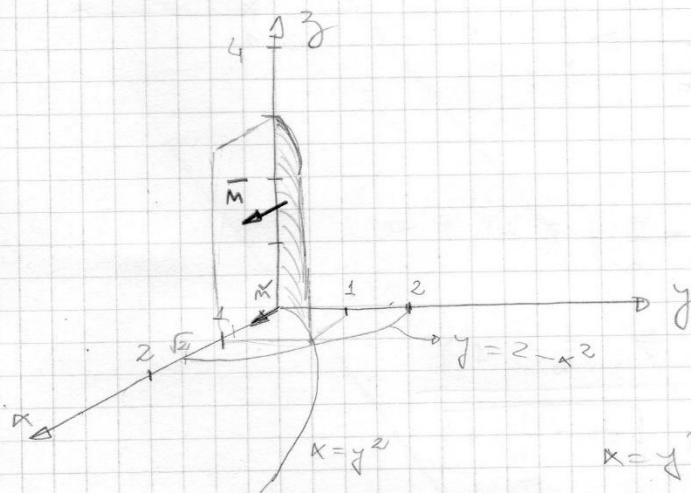


11)

$$\begin{cases} x = y^2 \\ y = 2 - x^2 \end{cases} \Rightarrow y = 2 - y^4 = 0$$

$$\Rightarrow y^4 + y - 2 = 0 \quad \begin{matrix} y=1 \\ y^3 + y^2 + y + 2 = 0 \end{matrix}$$

$$\begin{array}{c|cccccc} 1 & 0 & 0 & 1 & -2 & \\ \hline 1 & 1 & 1 & 1 & 2 & 0 \end{array}$$



$$x = y^2 \Rightarrow |y| = \sqrt{x} \quad (x \geq 0)$$

$$y = \sqrt{x} \quad (\text{since } y \geq 0)$$

$$F(x, y, z) = x - y^2 = 0$$

$$\vec{n} = \frac{\nabla F}{\|\nabla F\|} = \frac{(1, -2y, 0)}{\sqrt{1+4y^2}}$$

$$\iint_S \vec{n} \cdot d\vec{\sigma} = \iint_S (x - y, x - z, 0(x, y, z)) \cdot \frac{(1, -2y, 0)}{\sqrt{1+4y^2}} d\sigma =$$

$$= \iint_R \frac{(x - y - 2xy + 2yz)}{\sqrt{1+4y^2}} \cdot \frac{\sqrt{1+4y^2}}{|1-2y|} dx dz$$

$$= \int_0^1 dx \cdot \int_0^4 dz \cdot \frac{x - y - 2xy + 2yz}{2y} dx dz =$$

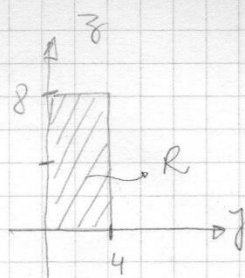
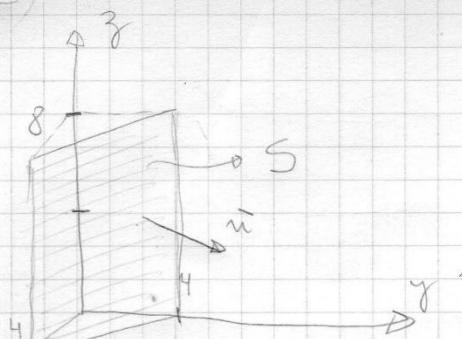
$$= \frac{1}{2} \int_0^1 dx \int_0^4 dz \frac{x - \sqrt{x} - 2x\sqrt{x} + 2\sqrt{x} - 3}{\sqrt{x}} =$$

$$= \frac{1}{2} \int_0^1 dx \int_0^4 (\sqrt{x} - 1 - 2x + 2 - 3) dz =$$

$$= \frac{1}{2} \int_0^1 \left[(\sqrt{x} - 1 - 2x) z + \frac{2}{2} z^2 \right]_0^4 dx =$$

$$= \frac{1}{2} \int_0^1 (4\sqrt{x} - 4 - 8x + 16) dx = \frac{1}{2} \left[4 \cdot \frac{2}{3} x^{3/2} - 4x - 8 \frac{x^2}{2} + 16x \right]_0^1 = \frac{1}{2} \left[\frac{8}{3} - 4 - 4 + 16 \right] = \frac{1}{2} \cdot \frac{16}{3} = \frac{16}{3}$$

12)



$$F(x, y, z) = x + y - 4 = 0$$

$$\vec{n} = \frac{\nabla F}{\|\nabla F\|} = \frac{(1, 1, 0)}{\sqrt{2}}$$

$$\iint_S \vec{n} \, d\sigma = \iint_S \nabla f \cdot \vec{n} \, d\sigma = \iint_S [1 + g'(x-y)] \, z;$$

$$[1 - g'(x-y)] \, z; g(x-y)] \cdot \frac{(1, 1, 0)}{\sqrt{2}} \, d\sigma =$$

$$= \iint_R \frac{1 + \cancel{g'(x-y)} + 1 - \cancel{g'(x-y)}}{\sqrt{2}} \, dy \, dz =$$

$$= \int_0^4 dy \int_0^8 dz \, \frac{2}{\sqrt{2}} = 2 \cdot 4 \cdot 8 = 64$$

13)

$$F(x, y, z) = y^2 - x^2 - z = 0$$

$$\vec{n} = \frac{\nabla F}{\|\nabla F\|} = \frac{(-2x, 2y, -1)}{\sqrt{4x^2 + 4y^2 + 1}}$$

$$\iint_S \nabla g \cdot \vec{n} \, d\sigma = \iint_S [y h'(xy), x h'(xy), 1] \cdot \frac{(-2x, 2y, -1)}{\sqrt{4x^2 + 4y^2 + 1}} \, d\sigma =$$

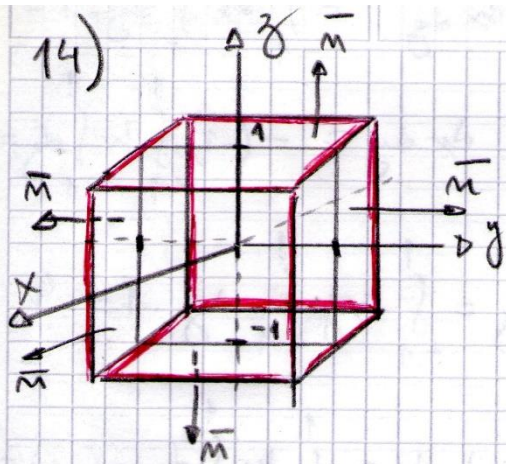
$$= \iint_D \frac{(-2xy h'(xy) + 2xy h'(xy) + 1)}{\sqrt{4x^2 + 4y^2 + 1}} \cdot \frac{\sqrt{4x^2 + 4y^2 + 1}}{|1-1|} \, dx \, dy$$

$$= \iint_D 1 \, dx \, dy = \iint_D dx \, dy = \text{area } D$$

14)

FECHA

Consideramos los seis
caras del cubo.



$$C_1 \rightarrow -1 \leq x \leq 1 \wedge -1 \leq y \leq 1 \wedge z = 1$$

$$\vec{n}_1 = \vec{k}$$

$$C_2 \rightarrow -1 \leq x \leq 1 \wedge -1 \leq y \leq 1 \wedge z = -1$$

$$\vec{n}_2 = -\vec{k}$$

$$C_3 \rightarrow -1 \leq x \leq 1 \wedge -1 \leq z \leq 1 \wedge y = 1$$

$$\vec{n}_3 = \vec{j}$$

$$C_4 \rightarrow -1 \leq x \leq 1 \wedge -1 \leq z \leq 1 \wedge y = -1$$

$$\vec{n}_4 = -\vec{j}$$

$$C_5 \rightarrow -1 \leq y \leq 1 \wedge -1 \leq z \leq 1 \wedge x = 1$$

$$\vec{n}_5 = \vec{i}$$

$$C_6 \rightarrow -1 \leq y \leq 1 \wedge -1 \leq z \leq 1 \wedge x = -1$$

$$\vec{n}_6 = -\vec{i}$$

$$\iint_S \vec{F} \cdot \vec{n} \, d\sigma = \iint_{C_1} + \iint_{C_2} + \dots + \iint_{C_6}$$

$$\bullet \iint_{C_1} (x, y, z) \cdot \vec{k} \, d\sigma = \iint_{C_1} z \, d\sigma = \iint_{D_1} z \, dx \, dy = \iint_{D_1} 1 \, dx \, dy =$$

$$= \int_{-1}^1 dx \int_{-1}^1 dy = \int_{-1}^1 dx [y]_{-1}^1 = \int_{-1}^1 2 \, dx = 2 \cdot 2 = 4$$

$$\bullet \iint_{C_2} (x, y, z) \cdot (-\hat{k}) d\sigma = \iint_{D_2} -z d\sigma = - \iint_{D_2} (-1) dx dy = \int_{-1}^1 dx \int_{-1}^1 dy =$$

NOTA $= 4$

$$\bullet \iint_{C_3} (x, y, z) \cdot \hat{j} d\sigma = \iint_{D_3} y dx dz = \int_{-1}^1 dx \int_{-1}^1 dz \cdot 1 = 4$$

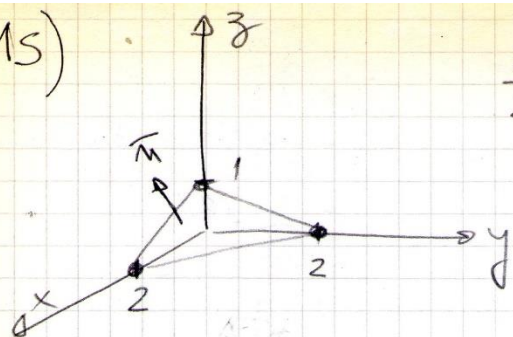
$$\bullet \iint_{C_4} (x, y, z) \cdot (-\hat{j}) d\sigma = \iint_{D_4} (-y) dx dz = -(-1) \int_{-1}^1 dx \int_{-1}^1 dz = 4$$

$$\bullet \iint_{C_5} (x, y, z) \cdot \hat{i} d\sigma = \iint_{D_5} x dy dz = \int_{-1}^1 dy \int_{-1}^1 dz \cdot 1 = 4$$

$$\bullet \iint_{C_6} (x, y, z) \cdot (-\hat{i}) d\sigma = \iint_{D_6} -x dy dz = \int_{-1}^1 dy \int_{-1}^1 dz - (-1) = 4$$

$$\iint_C (x, y, z) d\sigma = 4 \cdot 6 = 24$$

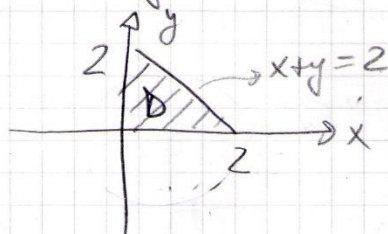
15)



$$\frac{x}{2} + \frac{y}{2} + \frac{z}{1} = 1$$

$$x + y + 2z = 2$$

$$F(x, y, z) = x + y + 2z - 2 = 0$$

Proj_{xy} (xy)

$$\vec{n} = \frac{\nabla F}{\|\nabla F\|} = \frac{(1, 1, 2)}{\sqrt{6}}$$

$$\iint_S \vec{f} \cdot \vec{n} \, d\sigma = \iint_S (x, y, z) \cdot \frac{(1, 1, 2)}{\sqrt{6}} \, d\sigma =$$

$$= \iint_S \frac{x + y + 2z}{\sqrt{6}} \, d\sigma = \iint_D \frac{x + y + 2z}{\sqrt{6}} \frac{\sqrt{6}}{|2|} \, dx \, dy =$$

$$= \frac{1}{2} \int_0^2 dx \int_0^{2-x} dy [x + y + 2(2 - x - y)] =$$

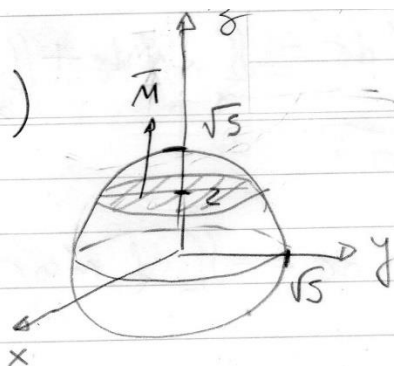
$$= \frac{1}{2} \int_0^2 dx \int_0^{2-x} dy [2 + y] = \frac{1}{2} \int_0^2 dx \left[2y + \frac{y^2}{2} \right]_0^{2-x} =$$

$$= \frac{1}{2} \int_0^2 dx \left[2(2-x) + \frac{(2-x)^2}{2} \right] =$$

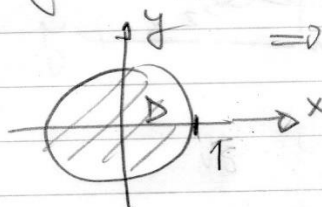
$$= \frac{1}{2} \left[4x - x^2 - \frac{(2-x)^3}{6} \right]_0^2 = \frac{1}{2} \left[8 - 4 - 0 + \frac{8}{6} \right] =$$

$$= \frac{8}{3}$$

16)



$$\text{Proj } \vec{r}(xy) : \left. \begin{aligned} x^2 + y^2 + z^2 &= 5 \\ z &= 2 \end{aligned} \right\} \Rightarrow x^2 + y^2 = 1$$



$$\iint_S \vec{F} \cdot \vec{n} \, d\sigma = \iint_S (x^2 z, z^3, x^2 z) \cdot (0, 0, 1) \, d\sigma =$$

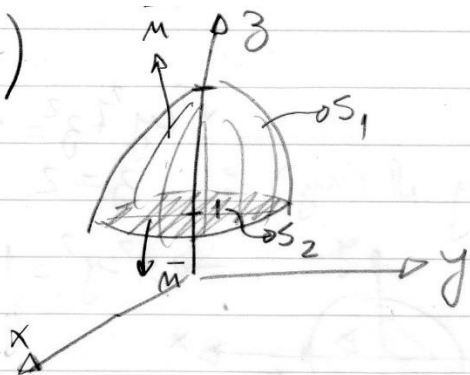
$$= \iint_S x^2 z \, d\sigma = \iint_D x^2 z \, dx \, dy =$$

$$= \iint_D x^2 \cdot 2 \, dx \, dy = 2 \int_0^{2\pi} d\theta \int_0^1 \rho \rho^2 \cos^2 \theta =$$

$$= 2 \int_0^{2\pi} \cos^2 \theta \, d\theta \left[\frac{\rho^4}{4} \right]_0^1 = 2 \cdot \frac{1}{4} \cdot \left[\frac{\theta}{2} + \frac{\sin(2\theta)}{4} \right]_0^{2\pi} =$$

$$= \frac{1}{2} [\pi] = \frac{\pi}{2}$$

17)

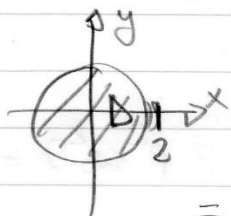


$$\underbrace{\iint_S \vec{f} \cdot \vec{n} \, d\sigma = \iint_{S_1} \vec{f} \cdot \vec{n} \, d\sigma + \iint_{S_2} \vec{f} \cdot \vec{n} \, d\sigma}_{= 0}$$

$$\iint_{S_1} \vec{f} \cdot \vec{n} \, d\sigma = - \iint_{S_2} \vec{f} \cdot \vec{n} \, d\sigma$$

Para S_2 proyector (x, y) $\left. \begin{array}{l} z = 5 - x^2 - y^2 \\ z = 1 \end{array} \right\} \Rightarrow$

$$\Rightarrow x^2 + y^2 = 4$$



$$- \iint_{S_2} \vec{f} \cdot \vec{n} \, d\sigma = - \iint_{S_2} \vec{f}(0, 0, -1) \, d\sigma =$$

$$= - \iint_{S_2} 3z \, d\sigma = \iint_D 3 \cdot 1 \, dx \, dy =$$

$$= 3 \underbrace{\int_0^{2\pi} d\theta \int_0^2 \rho \, d\rho}_{\text{área círculo}} = 3 \cdot \pi \cdot 2^2 = \boxed{12\pi}$$